

plan today: finish "closest point property" discussion
 direct sums, bounded operators, operator norm,
 dual space, Riesz representation theorem, adjoint (part 1)

last time: If $e_1, \dots, e_N$ is an O.N. basis for $V_0 \subseteq V$
 the orthogonal projection $P: V \rightarrow V_0$ is given by

we showed: 1. P is linear
 2. $\langle Pu, v \rangle = \langle u, Pv \rangle$
 3. If $v_0 \in V_0$, $Pv_0 = v_0$
 4. If $u \in V$, $v \in V_0$ then $\langle u - Pu, v \rangle = 0$

$$Pu = \sum_{n=1}^N \langle u, e_n \rangle e_n$$

Theorem:

$$\langle u - u_0, v \rangle = 0 \quad \forall v \in V_0 \quad (\Leftrightarrow) \quad \|u - u_0\| = \min_{v_0 \in V_0} \|u - v_0\|$$

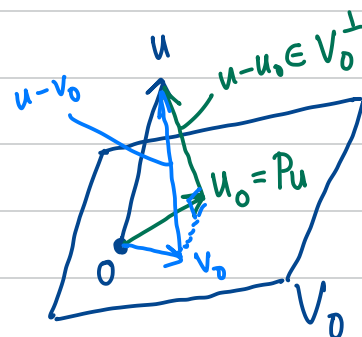
(so $u_0 = Pu$ is the closest point to u in V_0)

Proof: $\Rightarrow$ Suppose u_0 satisfies LHS

for any $v_0 \in V_0$, $\underbrace{u_0 - v_0}_v \in V_0$ so $\langle u - u_0, \underbrace{u_0 - v_0}_v \rangle = 0$

$$\begin{aligned} \therefore \|u - v_0\|^2 &= \|u - u_0 + u_0 - v_0\|^2 \\ &= \|u - u_0\|^2 + \|u_0 - v_0\|^2 \\ &\geq \|u - u_0\|^2 \end{aligned}$$

↪ with equality iff $u_0 = v_0$



$\Leftarrow$ We will show the contrapositive: $\text{not } A \Rightarrow \text{not } B$
(the book's proof of Cauchy-Schwarz is similar)

(1) Suppose $\exists v \in V_0$ s.t. $\langle u - u_0, v \rangle = r e^{i\theta} \neq 0$

Then $w = v e^{i\theta}$ satisfies

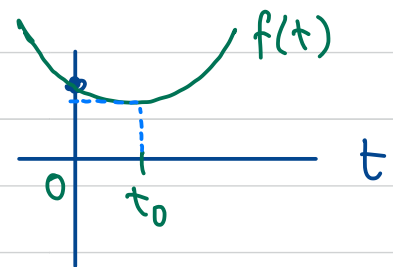
$$\langle u - u_0, w \rangle = e^{-i\theta} \langle u - u_0, v \rangle = r > 0$$

Since r is real, $\langle w, u - u_0 \rangle = r$ as well

(2) The function $f(t) = \|u - u_0 - t w\|^2$
 $= \langle u - u_0 - t w, u - u_0 - t w \rangle$
 $= \|u - u_0\|^2 - 2tr + t^2 \|w\|^2$

is differentiable and satisfies $f'(0) = -2r < 0$

$\therefore f(t)$ does not have a
minimum at $t = 0$



(3) There is a $t_0 > 0$ so that $f(t_0) < f(0)$

$$f(t_0) = \|u - u_0 - t_0 w\|^2 = \|u - z_0\|^2 < \|u - u_0\|^2 = f(0)$$

$$z_0 = u_0 + t_0 w \in V_0$$

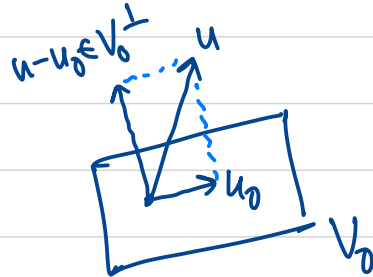
$$\text{So } \|u - u_0\| \neq \min_{v_0 \in V_0} \|u - v_0\|$$

Corollary : $V = V_0 \oplus V_0^\perp$ (assuming V_0 is finite-dimensional or, more generally, closed)
 direct sum

This means every $u \in V$ can be written uniquely as

$$u = u_0 + \underbrace{(u - u_0)}_{\substack{\text{orthogonal} \\ \text{to everything} \\ \text{in } V_0}}$$

$\nearrow$
 closest point
 to u in V_0
 $(u_0 = P u)$



Uniqueness : we showed $\|u - v_0\| > \|u - u_0\|$ unless $v_0 = u_0$
 (There is only one closest point.) $\nearrow$ strict

Norms of operators Let X, Y be normed vector spaces.

A linear operator $A: X \rightarrow Y$ is bounded if there is a constant M such that

$$\|Ax\|_Y \leq M \|x\|_X \quad \forall x \in X$$

$\nearrow$ norm in Y $\nearrow$ norm in X

The smallest such constant M is the operator norm of A :

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|=1} \|Ax\|$$

sup means supremum, or least upper bound.
(think of it as the maximum value)

To show that $\|A\| = M$ (e.g. in HW 2)

step 1. show that $\|Ax\| \leq M \|x\| \quad \forall x$ (M works)

step 2. either (a) produce an $x_0 \neq 0$ s.t. $\|Ax_0\| = M \|x_0\|$
or (b) show that if $K < M$ then $\|Ax_0\| > K \|x_0\|$ equivalently, $\geq$

(just to match $x \neq 0$ in the sup formula) $\exists x_0 \neq 0$ s.t. $\|Ax_0\| > K \|x_0\|$ ($K < M$ does not work)

Sometimes (a) is impossible. (a) or (b) shows that M is the least upper bound.

Theorem: If A is an $m \times n$ matrix,

$$(1) \|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |A_{ij}|$$

maximum absolute column sum
("1 is standing up")

$$(2) \|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|$$

maximum absolute row sum
("∞ is lying down")

$$(3) \|A\|_2 = \sqrt{\lambda_{\max}(A^*A)}$$

A^* = adjoint =
conjugate transpose

$\lambda_{\max}$ = largest eigenvalue

proof of ①: Let $M = \max_{1 \leq j \leq n} \sum_{i=1}^m |A_{ij}|$

step 1: Let $x \in \mathbb{C}^n$. Then

$$\begin{aligned} \|Ax\|_1 &= \sum_{i=1}^m |(Ax)_i| = \sum_{i=1}^m \left| \sum_{j=1}^n A_{ij} x_j \right| \\ &\leq \sum_{i=1}^m \left(\sum_{j=1}^n |A_{ij}| |x_j| \right) \quad \leftarrow \text{triangle inequality} \\ &= \sum_{j=1}^n \left(\sum_{i=1}^m |A_{ij}| \right) |x_j| \leq M \sum_{j=1}^n |x_j| = M \|x\|_1 \end{aligned}$$

step 2a: pick k so $\sum_{i=1}^m |A_{ik}| = M$

$$\text{Let } x_0 = e_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow k\text{th slot} \quad \left((x_0)_j = \delta_{kj} \right)$$

$$\text{Then } (Ax_0)_i = \sum_{j=1}^n A_{ij} (x_0)_j = A_{ik} \quad \|x_0\|_1 = \sum_{j=1}^n |\delta_{kj}| = 1$$

$$\text{and } \|Ax_0\|_1 = \sum_{i=1}^m |A_{ik}| = M = M \|x_0\|_1$$

$\therefore$ the bound is achieved and $\|A\|_1 = M$.

you'll prove ② in HW 2.

③ requires learning about the symmetric eigenvalue problem and the SVD.

dual spaces

Let V be a normed vector space. The dual space V'

is the space of bounded linear functionals on V ,

$$V' = \{ l: V \rightarrow \mathbb{C}, \text{ } l \text{ linear and bounded} \}$$

If V is an inner product space and $v \in V$, we define

$$l_v \in V' \text{ by } l_v(u) = \langle u, v \rangle.$$

Theorem: $\|l_v\| = \|v\|$.

proof: Let $M = \|v\|$, $u \in V$.

$$\text{step 1: } |l_v(u)| = |\langle u, v \rangle| \stackrel{\text{C.S.}}{\leq} \|u\| \cdot \|v\| = M\|u\| \quad \checkmark$$

step 2: Use method (a). Two cases:

If $v \neq 0$, let $u_0 = v$. Then

$$|l_v(u_0)| = |\langle v, v \rangle| = \|v\|^2 = M\|u_0\| \quad \checkmark$$

If $v = 0$, let u_0 be any nonzero vector in V .

$$|l_v(u_0)| = |\langle u_0, v \rangle| = |\langle u_0, 0 \rangle| = 0 = M\|u_0\| \quad \checkmark$$

$\uparrow$
 $M = \|v\| = 0$

def: A Hilbert space is a complete inner product space.
↑ Cauchy sequences converge

Riesz representation theorem: If X is a Hilbert space and $\ell \in X'$, there is a unique $v \in X$ s.t. $\ell = \ell_v$,
i.e. $\ell(u) = \langle u, v \rangle \quad \forall u \in X$.

so $\{\ell_v : v \in X\}$ is the entire dual space.

Adjoint of an operator: Let X, Y be inner product spaces

Suppose $A: X \rightarrow Y$ is a linear operator

we define: $A^*: Y \rightarrow X$ by requiring

$$(1) \quad \langle Ax, y \rangle_Y = \langle x, A^*y \rangle_X \quad \text{for all } x \in X, y \in Y$$

Equivalent condition:

$$(2) \quad \langle A^*y, x \rangle_X = \langle y, Ax \rangle_Y \quad \forall x \in X, y \in Y$$

The equivalence comes from $z = w$ iff $\bar{w} = \bar{z}$

$$\left[\begin{array}{l} \text{conjugate both sides of (1)} \\ \text{use } \overline{\langle Ax, y \rangle_Y} = \langle y, Ax \rangle_Y \\ \text{and } \overline{\langle x, A^*y \rangle_X} = \langle A^*y, x \rangle_X \end{array} \right.$$

$\therefore$ moving A or A^* to the other slot
converts it to A^* or A , respectively

Example: let $A: \mathbb{C}^n \rightarrow \mathbb{C}^m$ be an $m \times n$ matrix
 what matrix B is its adjoint?

solution: $B: \mathbb{C}^m \rightarrow \mathbb{C}^n$ satisfies

$$(By, x)_{\mathbb{C}^n} = (y, Ax)_{\mathbb{C}^m} \quad \forall x, y$$

$$\bar{x}^T By = \sum_{p=1}^n \bar{x}_p \left(\sum_{q=1}^m B_{pq} y_q \right)$$

can pick out the j th entry with $x = e_j, y = e_k$
 $(x_p = (e_j)_p = \delta_{jp}, y_q = \delta_{kq})$

$$\begin{aligned} B_{jk} &= e_j^T B e_k = (B e_k, e_j) = (e_k, A e_j) \\ &= \overline{(A e_j, e_k)} = \overline{e_k^T A e_j} = \overline{A_{kj}} \end{aligned}$$

$$A_{jk}^* = \overline{A_{kj}} \quad (\text{adjoint} = \text{conjugate transpose})$$

→ alternatively, just manipulate (y, Ax) until it has
 the form (By, x) :

$$\begin{aligned} (y, Ax)_{\mathbb{C}^m} &= \sum_{k=1}^m y_k \overline{(Ax)_k} = \sum_{k=1}^m \overline{\left(\sum_{j=1}^n A_{kj} x_j \right)} y_k \\ &= \sum_{j=1}^n \underbrace{\left(\sum_{k=1}^m \overline{A_{kj}} y_k \right)}_{(By)_j} \bar{x}_j = (By, x)_{\mathbb{C}^n} \end{aligned}$$

$B_{jk} = \overline{A_{kj}}$

next time we'll see that A^* exists, is unique,
and is itself a bounded linear operator with the
same norm as A . (assuming X is a Hilbert space)

Short comment about HW2 #5 (book p. 36 #15)

$$\psi(x) = \begin{cases} 1 & 0 \leq x < 1/2 \\ -1 & 1/2 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\psi(2x) = \begin{cases} 1 & a \leq x < b \\ -1 & b \leq x < c \\ 0 & \text{o.w.} \end{cases}$$

$$\psi(2x-1) = \begin{cases} 1 & d \leq x < e \\ -1 & e \leq x < f \\ 0 & \text{o.w.} \end{cases}$$

part of the
task is to
find a, b, c, d, e, f

e.g. when is
 $2x-1 \in [1/2, 1)$?